CONTROL SYSTEMS 354 · TEST

Question Paper

Total **80 marks** Questions **3** Memo discussed **9 Oct 2026**

> Reconstructed from the lecturer's memo walkthrough. The plants, specifications and mark allocations come from the recording; the wording of each question is a reconstruction, not the original text.

INSTRUCTIONS

1. Answer all questions.
2. Work from first principles where asked, and show all working.
3. The formula sheet supplies the settling time relations, the percentage overshoot against phase margin figure, and the maximum phase lead against $1/\alpha$ figure.

Question 1 · Open loop and PID selection [26]

Consider the plant

$$G(s) = \frac{5}{s^2 + 2s + 15}$$

The step response of the system must satisfy both of the following requirements:

(1) zero steady-state error to a unit step reference, so that $y_{ss} = 1$;

(2) a 2% settling time of $T_s < 1\ \text{s}$.

(a) The plant is used in an open-loop configuration with no control. [12]

(i) Using the final value theorem, determine the steady-state output y_{ss} for a unit step input. State whether requirement (1) is met. (6)

(ii) Determine the poles of $G(s)$. Sketch them on the s-plane together with the region in which poles must lie to satisfy requirement (2). Use your sketch to state whether requirement (2) is met. (6)

(b) A proportional gain K is placed in series with the plant, still in an open-loop configuration. Find the value of K that satisfies requirement (1), and state the main disadvantage of achieving the requirement with open-loop control. [4]

(c) The plant is now placed in a unity negative feedback loop. Propose a controller made up of some combination of proportional, integral and derivative action that will meet both requirements. Justify your choice from first principles, using system type and the steady-state error for reference tracking. You do not need to calculate the controller gains. [10]

Question 2 · Frequency response design [26]

The open-loop Bode plot of an uncompensated plant $G(s)$ is given in Figure 2.1.



Figure 2.1 Uncompensated open-loop Bode plot (magnitude and phase). The figure itself was handed out in the test and is not in the recording. From the memo, the plant is type 1 with $\lim_{s\to 0} sG(s) = 1$, and near $\omega \approx 14$ rad/s its magnitude is about -26 dB with phase close to $-180°$.

(a) Sketch the Bode magnitude and phase of a PD controller. Use your sketch to explain the benefits of PD control for the transient [6]

response and the relative stability of a closed-loop system.

(b) Using the frequency response design method and Figure 2.1, design a lead compensator of the form [20]

$$D(s) = K\,\frac{Ts+1}{\alpha Ts+1}, \qquad 0<\alpha<1$$

so that the unity feedback system meets both of the following:

(1) steady-state error $e_{ss} \le 0.05$ for a unit ramp input;

(2) percentage overshoot $\%OS < 5\%$ for a unit step input.

State the final compensator transfer function with all values substituted.

Question 3 · Root locus design [28]

The plant below is placed in a unity negative feedback loop. Use the root locus method throughout.

$$G(s) = \frac{1}{s(s+2)}$$

The closed-loop system must satisfy both of the following:

(1) dominant closed-loop poles at the design point $s_0 = -1 \pm j2$ (transient requirement);

(2) a velocity error constant $K_v = 100$, so $e_{ss} = 0.01$ for a unit ramp input (steady-state requirement).

3.1 Using the angle condition, show that the root locus passes through s_0, so that proportional control alone can meet requirement (1). State the transfer function $L(s)$ for which the root locus is drawn, and show how it follows from the characteristic equation. [8]

3.2 Using the magnitude condition, find the proportional gain K that places the closed-loop poles at s_0. Hence determine whether proportional control alone meets requirement (2). [10]

3.3 Design a lag compensator of the form [10]

$$D(s) = K\,\frac{s+z}{s+p}, \qquad \alpha = \frac{z}{p} > 1$$

so that both requirements are met. Justify the placement of the pole and zero, and state the final compensator transfer function.

End of paper Total 80